

GAGAN JAIN

Software Engineer · Full Stack & AI/LLM Developer

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PROFESSIONAL SUMMARY

Systems and full-stack engineer who built a programming language from first principles in Rust and production-style web apps in Python and JavaScript. Hands-on experience includes layered Flask/MySQL architecture, authentication and security hardening, CI/CD, Docker deployment, monitoring, REST APIs, WebSockets, and server-side route protection. Focused on maintainable web systems, debugging, and deployment hygiene for entry-level software roles.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, C, C++, HTML, CSS

Backend & APIs: Flask, Node.js, Express, Laravel, Axum (Rust), REST APIs, WebSockets, MySQL, MongoDB Atlas

Frontend: React, React Native, Tailwind CSS, Next.js, Monaco Editor

DevOps & Tooling: Docker, Git, GitHub Actions, Linux, Prometheus, CI/CD Pipelines

AI / ML: Ollama, YOLO v8, Computer Vision, NLP, Transformer Architecture Research

Testing: pytest, Jest, Integration Testing (207 tests across projects)

Methodologies: Agile, Test-Driven Development (TDD), Layered Architecture, Code Review

PROJECTS

Vyākṛti — Programming Language & Compiler IDE — Rust, React, TypeScript, Axum | github.com/gaganjainse/Vyakrti

- Designed and implemented a Sanskrit-keyword language from scratch: 38+ token-type lexer, recursive-descent parser, and a stack-based bytecode VM with 34 opcodes — all hand-written in Rust without parser-generator tooling.
- Built a real-time web IDE (React + Monaco Editor) backed by Rust/Axum REST API and WebSocket endpoints for live REPL execution and remote file management.
- Achieved self-hosting: the compiler successfully parses and compiles its own source code — a standard benchmark for programming language maturity.
- Validated across 123 integration tests covering lexer tokenisation, parser AST correctness, and VM opcode execution; full language spec documented.

AIM — Attendance Information Manager — Python, Flask, MySQL, Docker, CI/CD | github.com/gaganjainse/AIM

- Built a production-grade web platform with strict 3-layer separation (API routes, domain services, data repositories) to simplify testing and maintenance.
- Implemented defense-in-depth security: Argon2id password hashing, CSRF protection, brute-force lockouts, breached-password scanning, JWT session tokens, and strict CSP/HSTS headers.
- Deployed Prometheus metrics, structured JSON logging, Chart.js analytics, FullCalendar scheduling views, and GitHub Actions CI/CD pipeline; packaged and run via Docker.
- Shipped with 84 automated pytest tests covering authentication flows, API contracts, and edge cases; integrated role-based access control for admin and regular users.

FWRS — Food Waste Reduction System — Python, Flask, Linear Programming, Folium | github.com/gaganjainse/FWRS

- Built an optimization platform that allocates surplus food from restaurants to NGOs using multi-stage linear programming, minimizing waste while maximizing social impact.
- Implemented 3-stage lexicographic solver prioritizing fairness → NGO priority → transport cost; expiry-aware routing penalizes allocations where travel time exceeds food shelf life.
- Developed interactive web dashboard with Flask and desktop GUI with Tkinter; features include Folium/Leaflet maps with animated routes, heatmap visualization, and priority-colored markers.
- System processes 100+ restaurants and 80+ NGOs with priority-weighted distribution (1-5 scale); exports allocations and summary metrics as CSV files for government/NGO reporting.

GameVault — Social Gaming Marketplace — Next.js 13, TypeScript, Supabase, Tailwind CSS |

github.com/gaganjainse/GameVault

- Built a social gaming marketplace for discovering and buying games, following creators, posting to a feed, watching livestreams and video uploads, and reselling owned digital assets.
- Implemented Supabase/Postgres auth with row-level security, server-side route protection in middleware, and SSR clients so protected pages are gated before HTML ships.
- Kept checkout integrity server-verified through Postgres-side logic; counters and feed signals are driven by database rules and triggers instead of client-side state.

EXPERIENCE

CodenPlay Robotics, Bengaluru | *Software & Automation Engineer Intern* | Aug 2025 – Nov 2025

- Programmed PLC control logic for automated production workflows using CODESYS structured text and ladder logic.
- Designed HMI monitoring dashboards and multi-device communication networks using Node-RED; exposed real-time process telemetry through flow-based interfaces.
- Validated automation sequences against Factory I/O digital-twin simulations prior to hardware deployment, reducing on-floor debugging cycles.

EDUCATION

Vellore Institute of Technology, Vellore — B.Tech in Computer Science and Engineering | 2026

CGPA: 7.5 / 10.0 | Relevant Coursework: Data Structures & Algorithms, Operating Systems, Compiler Design, Database Systems, Computer Networks